Venkata Lakshmi Parimala Pasupuleti

Sales Engineer · LangChain / LangGraph / LangSmith · Production AI Agent Delivery · Python & LLM Evaluation

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SUMMARY

AI engineer with 9 years of production software ownership and three independent LangGraph systems shipped in production. Built AgentProbe -- an open-source evaluation and observability framework for AI agents with a CI-integrated regression gate, LLM-as-judge scoring, and multi-provider routing -- solving the same architectural problem LangSmith solves, independently, because I needed it to operate LeaseI in production. Built LeaseI -- an 8-agent LangGraph platform for enterprise compliance operations with RAG, HITL guardrails, and structured data extraction, evaluated across 1,240 test cases at 93.6% precision. Deployed LangGraph for a third time at Magizhs, delivering an agentic planning system for an external client. Seven years at Entrata shipping production Python backend systems at scale. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, 2026).

SKILLS

LangChain Ecosystem: LangGraph, LangChain, LLM Evaluation, Observability for AI Agents, Agent Tracing, Regression Suites, CI Eval Gate, LLM-as-Judge, Pass@k Harness, Multi-Step Workflows, Agent Orchestration, Failure Handling, HITL, Guardrails, RAG Pipelines, Tool Calling, Function Calling, MCP (Model Context Protocol), Prompt Engineering, Output Validation

AI & LLM Providers: Anthropic API, OpenAI API, Groq, Ollama, ChromaDB, Vector Database, Semantic Search, Embeddings, Embedding Models, OCR, Document Intelligence, POC Development, Hallucination Detection, Structured Outputs

Languages & Frameworks: Python, JavaScript, TypeScript, Java, SQL, FastAPI, REST APIs, React.js, Angular, Node.js, Flask, HTML/CSS, Bash

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, Kubernetes (K8s), CI/CD (GitHub Actions), RabbitMQ, Redis, Microservices, Distributed Systems, Production Monitoring, Git, Linux, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern -- External Client Delivery

- Delivered a production LangGraph agentic system for an external client (parse → plan → retrieve → validate → conditional routing) -- owned design through delivery, including build-vs-buy decisions on orchestration, RAG, and validation layers.
- Ran architecture reviews and technical walkthroughs with client stakeholders; translated complex agent orchestration concepts into clear tradeoff explanations for non-specialist audiences.

Entrata (PropTech / Real Estate SaaS · 100K+ MAU · production Python backend)
Sep 2018 - Aug 2024

Senior Software Engineer

- Shipped 20+ full-stack production modules (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign using Docker (50% scalability improvement, 30% deployment time reduction); owned backend systems through design, development, QA, and release.
- Built and owned RabbitMQ event-driven async processing system with retry logic and idempotency guarantees; owned production incidents and annual change cycles for 7 years.
- Optimized 200+ SQL queries (30% performance gain); AWS S3 migration with 100% PHI compliance; eliminated 80% of manual operational workload through Python automation.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews; built internal debugging and tooling that reduced debugging time by 40%.

Entrata India (Db Xento Systems)
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee Software Engineer

- Built property management backend modules (Python, Java, PostgreSQL); AWS S3 migration; database architecture redesign migrating hundreds of thousands of records.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Delivered Python programming and database design workshops to 600+ students; built prediction models ($R^2 = 0.977$) and anomaly detection pipelines on multi-source research datasets.

PROJECTS

AgentProbe: LLM Evaluation & Observability Framework (LangSmith-equivalent)
2025 | github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Multi-provider LLM routing

- Built an open-source evaluation and observability platform for AI agents -- the same architectural problem LangSmith solves. 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate that blocks quality degradation on every

commit, LLM-as-judge scoring with multi-trial pass@k harness, agent tracing, and multi-provider LLM routing across Anthropic, OpenAI, Groq, and Ollama. 15-endpoint FastAPI backend.

- Built independently to evaluate Leasel -- proof that non-deterministic AI agents can be operated reliably in production only if you instrument and gate quality before deployment.

Leasel: 8-Agent LangGraph Production System

2026 | github.com/venkatalakshmiparimala/Lease_complaince_rag

Python · LangGraph (8-agent StateGraph) · FastAPI · RAG · OCR · ChromaDB · SQLite · Streamlit

- Deployed an 8-agent LangGraph StateGraph in production: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation. Conditional routing, HITL on regulated decisions, RAG over compliance documents, failure handling throughout. The exact architecture customers ask "how do I actually build this in production?"
- 93.6% precision, 87.9% F1, pass@1 81.7% across 1,240 test cases -- with full observability through AgentProbe eval harness.

AI on FHIR: NLP Medical Records Query System

2024 | github.com/venkatalakshmiparimala/AI_on_FHIR

Flask · React · LLMs · FHIR · BERT · Docker

- LLM + BERT + embedding integration in production Flask + React system; HIPAA-compliant RBAC; NLP-to-FHIR API translation reducing manual retrieval time ~60%.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — Graduated May 2026 | GPA: 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017